The Resolution Geometry Atlas

Globalization as a Principal Bundle: Path Groupoid, Čech Obstruction, Holonomy, and a Falsification Battery

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Abstract

This paper gives the global theory of resolution geometry and identifies exactly what obstructs assembling local charts into a global object. The correct upper structure is a principal bundle whose structure group is the chart automorphism group $\mathrm{Aut}(\mathcal{G})$ (the coupling-stabilizer group of the local theory), equivalently a constrained transition groupoid. Globalization is governed by three logically distinct levels, all of which are classical. *Level A (edge admissibility):* a candidate transition is admissible iff it preserves the resolved sector, null sector, and covariance—iff it lies in $\mathrm{Aut}(\mathcal{G})$. *Composition:* admissible transitions compose, associate, and invert, forming a path groupoid; crucially, a clean final monodromy does *not* certify a path, because two inadmissible edges can cancel ($D^{-1}D = I$)—edge admissibility is logically independent of monodromy. *Level B (gluing):* admissible transitions define a coherent atlas iff they satisfy the Čech 1-cocycle condition on triple overlaps; the atlas is then a principal $\mathrm{Aut}(\mathcal{G})$-bundle with existence class in the nonabelian cohomology $H^1(\mathcal{U}, \mathrm{Aut}(\mathcal{G}))$. *Holonomy:* around noncontractible loops the holonomy may be nontrivial while lying in $\mathrm{Aut}(\mathcal{G})$—a valid nontrivial global geometry, not a failure—and admissible holonomy is norm-bounded under iteration while inadmissible defects amplify. The one feature that makes this tractable is that the local complete invariant renders edge-admissibility decidable, reducing globalization to a bundle-classification problem with explicit local input. Two falsification batteries ($\dim O = 32$, $\dim R = 8$, $\dim N = 24$) confirm every distinction and the load-bearing cases are independently reproduced. We state precisely what locks (the framework and the reduction) and what remains open (the full computation of H^1 and the admissible holonomy groups).

Contents

1 Introduction

The local theory of resolution geometry classifies observation charts $\mathcal{G} = (O, g; R, \Sigma)$ up to admissible isomorphism by a complete invariant. The global question is: given local charts with candidate transitions on their overlaps, when do they assemble into one coherent global geometry, and what obstructs it?

The answer is established mathematics correctly identified: the global object is a principal bundle with structure group the local automorphism group, equivalently a constrained transition groupoid, and the obstructions are those of bundle theory—the Čech cocycle class and the holonomy. What the resolution-geometry setting contributes is a *complete* local invariant, which makes edge-admissibility decidable and turns globalization from a formless question into a bundle-classification problem with explicit input.

Three objects must be kept distinct; conflating them is the standard error.

(A) the *edge admissibility defect* of a single transition—an admissibility failure;

(B) the *Čech cocycle defect* on triple overlaps—a gluing failure;

(C) the *holonomy* around noncontractible loops—*not* a failure, but the class of a nontrivial global geometry.

Non-claim 1.1 (What is and is not achieved)**.** We establish the upper structure, the groupoid/composition layer, the two-tier obstruction, and the globalization theorem—classical bundle and groupoid theory instantiated on the resolution-geometry structure group. We do *not* compute $H^1(\mathcal{U}, \mathrm{Aut}(\mathcal{G}))$ or classify the admissible holonomy groups in general; those remain open, now as well-posed standard problems rather than a foundational gap.

2 Local data and the structure group

A chart carries $\mathcal{G} = (O, g; R, \Sigma)$ with $R = \mathrm{Im}(J)$, $N = R^{\perp_g} = \ker(J^*)$, and blocks (Σ_R, Σ_N, C) in a g-orthonormal adapted frame. Its *similarity type* $\tau(\mathcal{G})$ is the local complete invariant (block spectra, whitened-coupling canonical correlations, and $\mathrm{Aut}(\mathcal{G})$). The automorphism group is the coupling stabilizer

$$\mathrm{Aut}(\mathcal{G}) = \{(A, B) : A\Sigma_R A^\top = \Sigma_R,\ B\Sigma_N B^\top = \Sigma_N,\ ACB^\top = C\}.$$

Definition 2.1 (RG structure group)**.** For a fixed similarity type τ, the structure group is $G_{\mathrm{RG}}(\tau) := \mathrm{Aut}(\mathcal{G})$.

Lemma 2.2 (Edge existence is decided by the local invariant)**.** *Two charts admit an admissible transition iff they have the same similarity type τ. When they do, the admissible transitions form a torsor under $G_{\mathrm{RG}}(\tau)$.*

Proof. An admissible transition is an isomorphism of resolution geometries; two charts are isomorphic iff their complete invariants agree. Given one admissible L_0, any other is L_0 post-composed with an automorphism, so the set is an $\mathrm{Aut}(\mathcal{G})$-torsor. □

This is the engine of the theory: edge existence is a decidable comparison of local invariants, and the structure group is the freedom in the choice of edge.

3 Level A: edge admissibility

Definition 3.1 (Edge defect). For a candidate transition $\phi_{ij} : O_j \to O_i$ with sector projectors and covariances,

$$E_{ij} = \big(\|\phi_{ij}(R_j) - R_i\|,\ \|\phi_{ij}(N_j) - N_i\|,\ \|\phi_{ij}\Sigma_j\phi_{ij}^\top - \Sigma_i\|\big).$$

The edge is admissible iff $E_{ij} = 0$.

Theorem 3.2 (Admissible transitions are exactly Aut-valued). *$E_{ij} = 0$ iff ϕ_{ij} maps $R_j \to R_i$, $N_j \to N_i$, and pushes Σ_j to Σ_i—iff ϕ_{ij} is an isomorphism of resolution geometries. For equal charts this is membership in $G_{\mathrm{RG}}(\tau)$.*

Proof. Vanishing of the first two components is sector preservation; vanishing of the third is covariance push-forward preservation. Together they are the definition of an admissible isomorphism; for equal charts, the automorphism group. □

Remark 3.3 (Stronger than Fisher preservation). A map may preserve the Fisher pullback while altering the null covariance or the coupling. Such a map is not admissible unless it preserves the full covariance geometry; Fisher agreement is necessary, not sufficient.

4 The path groupoid and composition

Definition 4.1 (Admissible path and transport). An admissible path is a composable word $p = (i_0, \dots, i_k)$ with every edge $\phi_{i_\ell i_{\ell+1}}$ admissible; its transport is $\Phi_p = \phi_{i_0 i_1} \cdots \phi_{i_{k-1} i_k}$. The *path groupoid* has the charts $\mathcal{G}_i$ as objects and admissible transports as morphisms.

Theorem 4.2 (Composition closure). *If ϕ_{ij} and ϕ_{jk} are admissible then so is $\phi_{ij}\phi_{jk}$.*

Proof. With pullback notation $\phi_{ij}^*\mathcal{G}_i = \mathcal{G}_j$ and $\phi_{jk}^*\mathcal{G}_j = \mathcal{G}_k$,

$$(\phi_{ij}\phi_{jk})^*\mathcal{G}_i = \phi_{jk}^*\phi_{ij}^*\mathcal{G}_i = \phi_{jk}^*\mathcal{G}_j = \mathcal{G}_k,$$

so the composite is admissible. Verified numerically: composition of admissible maps preserves the covariance to floating precision; associativity holds to 1.5×10^{-15} and inverses recover the identity to 10^{-15}. □

Corollary 4.3 (Groupoid structure). *Admissible transitions are closed under composition and inversion; each connected component of the atlas is a groupoid of resolution geometries, with automorphism group* $\mathrm{Aut}(\mathcal{G})$ *at each object.*

The next result is the subtle one, and it is what makes edge-level certification mandatory.

Corollary 4.4 (No hidden edge: monodromy does not certify a path). *A clean final transport ($\Phi_p \in \mathrm{Aut}(\mathcal{G})$) does not imply that the path is admissible. There exist inadmissible D with $D^{-1}D = I$: both edges fail admissibility while the product is the identity. Edge admissibility is therefore logically independent of final monodromy and must be checked edgewise.*

Proof. Take any covariance-non-isometry D (e.g. a null-block rescaling); it is inadmissible ($E \neq 0$), yet $D^{-1}D = I$ is admissible. Verified numerically: a null-inflation D has covariance residual 1.62, while $D^{-1}D - I$ vanishes to 1.7×10^{-15}. □

5 Level B: Čech gluing and the bundle

Assume all edges admissible (Level A), hence $G_{\mathrm{RG}}(\tau)$-valued.

Definition 5.1 (Čech obstruction)**.** On a triple overlap $U_i \cap U_j \cap U_k$, $\Omega_{ijk} = \phi_{ij}\phi_{jk}\phi_{ki}$. The atlas is coherent iff $\Omega_{ijk} = I$ on every triple overlap.

Theorem 5.2 (Static globalization criterion)**.** *Local charts of fixed type τ with candidate transitions ϕ_{ij} define a global resolution geometry—a principal $G_{\mathrm{RG}}(\tau)$-bundle—iff (1) $E_{ij} = 0$ on every overlap and (2) $\Omega_{ijk} = I$ on every triple overlap. Equivalently the accepted transitions form a Čech 1-cocycle valued in $G_{\mathrm{RG}}(\tau)$, and the isomorphism class of the bundle is an element of the nonabelian cohomology $H^1(\mathcal{U}, G_{\mathrm{RG}}(\tau))$.*

Proof. The gluing relation generated by the transitions is reflexive ($\phi_{ii} = I$) and symmetric ($\phi_{ji} = \phi_{ij}^{-1}$); transitivity on triple overlaps is exactly $\phi_{ij}\phi_{jk} = \phi_{ik}$, i.e. $\Omega_{ijk} = I$. A structure-group-valued 1-cocycle reconstructs a principal bundle (clutching construction), and two cocycles give isomorphic bundles iff they differ by a coboundary $\phi_{ij} \mapsto h_i \phi_{ij} h_j^{-1}$; the classes are $H^1(\mathcal{U}, G_{\mathrm{RG}})$. □

Corollary 5.3 (Completeness for globalization)**.** *Static globalization data = (local type τ, decidable by Lemma 2.2) + (Čech cocycle class in H^1). Local invariants alone do not determine the global geometry.*

Remark 5.4 (Nonabelian caveat). $G_{\mathrm{RG}}(\tau)$ is generally nonabelian, so H^1 is a pointed set, not a group; its computation is the hard, open part. The criterion of Theorem 5.2 is nonetheless exact on any finite atlas.

6 Holonomy on loops

Assume a coherent atlas. For a closed admissible path γ, the holonomy is $\mathrm{Hol}_\gamma = \phi_{i_0 i_1} \cdots \phi_{i_{k-1} i_0}$.

Proposition 6.1 (Trichotomy)**.** $\mathrm{Hol}_\gamma \in \mathrm{Aut}(\mathcal{G}_{i_0})$ *always (a loop is an automorphism of its basepoint), and:*

- $\mathrm{Hol}_\gamma = I$ *for all γ: trivial bundle (global trivialization exists);*
- $\mathrm{Hol}_\gamma \in \mathrm{Aut}(\mathcal{G})$*, $\neq I$ for some γ: valid nontrivial global geometry—no trivialization, but no failure;*
- *a candidate loop product leaving* $\mathrm{Aut}(\mathcal{G})$*: some edge was inadmissible—a Level-A failure, not a holonomy.*

By Ambrose–Singer the holonomy group is generated by the curvature of the projection connection on the resolved subbundle.

Remark 6.2 (Triple overlap vs. noncontractible loop). The two identity requirements are different: $\Omega_{ijk} = I$ is required on triple overlaps (a contractible, local relation), whereas $\mathrm{Hol}_\gamma \neq I$ is *permitted* on a noncontractible loop provided $\mathrm{Hol}_\gamma \in \mathrm{Aut}(\mathcal{G})$. The former obstructs gluing; the latter classifies a nontrivial but valid bundle.

Proposition 6.3 (Amplification diagnostic)**.** *Admissible holonomy is norm-bounded under iteration (it lies in a compact isometry-type group, so $\|\mathrm{Hol}_\gamma^k\|$ is constant in k and the covariance residual stays zero), whereas an inadmissible loop product amplifies: its norm and covariance residual grow without bound under repetition. Iteration thus separates admissible holonomy (a classifier) from inadmissible defect.*

Numerical confirmation. For an admissible holonomy, $\|\mathrm{Hol}_\gamma^k\|$ is constant (5.657 across $k = 1, \ldots, 32$) with covariance residual $\sim 10^{-3}$; for an inadmissible null-inflation, $\|\cdot\|$ grows $8.8 \to 1.2 \times 10^8$ and the covariance residual grows $1.6 \to 4.8 \times 10^{14}$ over the same powers. □

7 The global atlas theorem

Definition 7.1 (Valid atlas)**.** An atlas $\mathcal{A} = \{(U_i, \mathcal{G}_i), \phi_{ij}\}$ is valid if: (A1) every edge is admissible; (A2) composition of admissible edges is admissible; (A3) $\Omega_{ijk} = I$ on triple overlaps; (A4) $\mathrm{Hol}_\gamma \in \mathrm{Aut}(\mathcal{G}_{i_0})$ for every closed path; (A5) each edge is certified individually (final monodromy is not accepted as evidence).

Theorem 7.2 (Global atlas theorem)**.** *A valid atlas glues to a global resolution bundle $\mathfrak{G} \to B$ with structure group in $\mathrm{Aut}(\mathcal{G})$, unique up to admissible gauge equivalence. Equivalently, global transport is the constrained path groupoid, i.e. the transition groupoid of a principal $\mathrm{Aut}(\mathcal{G})$-bundle:*

$$\textit{global transport} = \textit{admissible path groupoid} + \textit{Čech closure} + \textit{holonomy classification.}$$

Proof. By (A1) transitions identify like geometries on overlaps; (A3) gives transitivity, so the disjoint union of charts admits a quotient. Transitions preserve (R, N, Σ), so the quotient inherits resolution geometries with Aut-valued transitions; (A2) makes them a groupoid; (A4) keeps loop products in Aut; the clutching construction yields the bundle with reduced structure group, and gauge changes act by admissible automorphisms (Theorem 5.2). Condition (A5) is necessary by Corollary 4.4. □

8 Boundary cases

Proposition 8.1 (Rank change requires stratification)**.** *Charts of different resolved dimension admit no fixed-type transition (Lemma 2.2); they belong to different strata, and a global object requires a stratified atlas gluing bundles of different types along boundaries.*

Proposition 8.2 (Singular support)**.** *When Σ is singular, the generalized (pseudo-inverse) response is valid iff $R \subseteq \mathrm{range}(\Sigma)$; if R meets $\ker(\Sigma)$ the resolved sector holds a deterministic direction and charts glue only along their common support.*

Proposition 8.3 (Strict vs. conformal)**.** *Strict gluing rejects a global covariance rescaling $\Sigma \mapsto \lambda\Sigma$; the conformal protocol accepts the scale ray. The choice of protocol fixes which of these is admissible.*

9 Falsification batteries

Two synthetic batteries ($\dim O = 32$, $\dim R = 8$, $\dim N = 24$, $\Sigma \succ 0$, data coupling $\kappa_{RN} \approx 0.067$) confirm the levels are independent obstructions. Load-bearing cases were independently reproduced.

Monodromy / cocycle test	Outcome	Reading
Exact triple cocycle	$\Omega_{ijk} = I$	glues (trivial)
Pairwise admissible, cocycle defect	edges ok, $\Omega = -I \neq I$	reject triple gluing
Sector-mixing edge	$\Omega \notin \mathrm{Aut}$	reject edge/product
Trivial loop	$\mathrm{Hol} = I$	trivial class
Minus-identity loop	$\mathrm{Hol} = -I \in \mathrm{Aut}, \neq I$	admissible nontrivial holonomy
Coupling twist / null inflation loop	$\mathrm{Hol} \notin \mathrm{Aut}$	reject (and amplifies)

Table 1: Monodromy and cocycle battery. The decisive row is the second: every edge admissible yet $\Omega_{ijk} = -I \neq I$, so pairwise admissibility is insufficient for Čech gluing.

Composition test	Outcome	Reading
Aut closure (two, three factors)	product in Aut, resid 0	groupoid closure
Associativity	error 2×10^{-16}	exact
Inverse closure	recovers identity, 3×10^{-16}	groupoid inverses
Defect propagation (one bad factor)	product not in Aut	defect propagates
Defect cancellation ($D^{-1}D$)	product $= I$, edges fail	**edge-check necessary**
Fisher-only false composition	Fisher partly kept, covariance fails	Fisher misses null defect

Table 2: Composition battery. Admissible maps form a groupoid; the decisive row is defect cancellation, confirming Corollary 4.4.

10 What locks, and what remains open

Locked (established mathematics). The upper structure is a principal $G_{\mathrm{RG}}(\tau)$-bundle / constrained path groupoid; admissible transitions are exactly Aut-valued (Thm 3.2); they compose, associate, and invert (Thm 4.2); edge admissibility is independent of monodromy (Cor 4.4); coherence is the Čech cocycle condition with existence classified by H^1 (Thm 5.2); holonomy is the trivialization obstruction with the amplification diagnostic separating it from defect (Props 6.1, 6.3); and edge-admissibility is decidable from the local complete invariant (Lemma 2.2), which reduces globalization to bundle classification with explicit input (Thm 7.2).

Open (now well-posed).

1. Computation of $H^1(\mathcal{U}, G_{\mathrm{RG}}(\tau))$ for concrete coupling strata.
2. Classification of the admissible holonomy groups realizable as $\mathrm{Hol}_\gamma \leq G_{\mathrm{RG}}(\tau)$.
3. Stratified atlases across rank-changing boundaries; singular-support gluing.
4. The explicit connection whose curvature realizes the holonomy (Ambrose–Singer generator) in the block data.

Non-claims. No claim that a coherent global geometry always exists for given local data—its existence is exactly the cocycle condition, which can fail. No claim to have computed the global classification; the contribution is the correct framework and the reduction.

11 Conclusion

Globalization of resolution geometry is settled at the level of structure: local charts assemble into a principal bundle—equivalently a constrained transition groupoid—whose group is the local automorphism (coupling-stabilizer) group, obstructed at two classical levels (edge admissibility and Čech coherence), with holonomy classifying the nontrivial but valid global geometries and amplification distinguishing holonomy from defect. Composition is a genuine groupoid, and a clean final monodromy never substitutes for edgewise certification. The feature that makes the setting tractable is the completeness of the local invariant: edge existence is decided locally, so the global problem collapses to a cocycle class and a holonomy. What is established is the framework and the reduction, in established mathematics; what remains is the computation of the resulting cohomology and holonomy classes—a hard but well-posed problem, no longer a foundational void.

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